operations in urban settings, industrial sites, forests, and confined indoor spaces.

Traditional obstacle avoidance relied primarily on ultrasonic and infrared sensors, which provided only short-range and limited-directional awareness. However, as drones began performing more complex missions, these basic systems proved inadequate. Modern designs now incorporate advanced sensor arrays including LiDAR (light detection and ranging), stereo vision cameras, millimeter-wave radar, and time-of-flight (ToF) sensors to create high-resolution, three-dimensional environmental maps. These sensors allow drones to detect objects of varying sizes and materials, even under poor lighting or weather conditions.

The integration of artificial intelligence (AI) and machine learning algorithms has significantly improved obstacle recognition and decision-making capabilities. Using techniques such as deep learning-based object detection and path prediction, drones can classify obstacles, anticipate movement, and plan optimal avoidance trajectories in real time. In swarm or coordinated operations, inter-drone communication systems further enhance safety by enabling drones to share environmental data and avoid collisions collectively.

Despite these advancements, challenges persist. Sensor performance can be affected by rain, fog, glare, or electromagnetic interference, and balancing computational demand with onboard power limitations remains a concern. Continuous improvements in edge computing, AI chip efficiency, and lightweight sensor integration are addressing these constraints, allowing faster data processing and more responsive flight control.

Ensuring robust and intelligent obstacle detection and avoidance is vital for achieving true autonomy in next-generation drones. With ongoing research in multi-sensor fusion, predictive modelling, and AI-enhanced